

Amylase

Test Details

METHODOLOGY

Enzymatic

RESULT TURNAROUND TIME

Within 1 day

Turnaround time is defined as the usual number of days from the date of pickup of a specimen for testing to when the result is released to the ordering provider. In some cases, additional time should be allowed for additional confirmatory or additional reflex tests. Testing schedules may vary.

Use

Work-up for abdominal pain, epigastric tenderness, nausea, and vomiting. Such findings characterize acute pancreatitis as well as acute surgical emergencies such as gastrointestinal perforation (eg, peptic ulcer with perforation) or bowel infarct. Amylase is used in the differential diagnosis of acute or chronic pancreatitis, which may or may not in an individual be related to alcoholism. Hypercalcemia related to pancreatitis is described with hyperparathyroidism and other entities. About 80% of subjects with acute pancreatitis have increased serum amylase within 24 hours.¹

Limitations

Poor specificity. Oxalate or citrate depress results. Lipemic sera (hypertriglyceridemia) may contain inhibitors which falsely depress results. About 20% of patients with acute pancreatitis have abnormal lipids. Normal serum amylase may occur in pancreatitis, especially relapsing and chronic pancreatitis. (Subjects in whom pseudocysts complicate chronic pancreatitis often do have elevations of the pancreatic enzymes.) The entire pancreas can be destroyed in pancreatitis; in such cases serum amylase will derive from other structures (eg, the salivary glands). Urine amylase increases often persist longer than do those of serum. High levels in alcoholics, in pregnancy and in diabetic ketoacidosis are of salivary rather than pancreatic origin. Salivary type amylase makes up about 60% of the enzyme, while it is the pancreatic fraction that is of clinical interest.² The expression “salivary amylase” includes other nonpancreatic sources of the enzyme. Serum amylase is cleared by renal excretion. Serum amylase may increase one to two times upper limit of normal in renal failure without diagnostic significance. In such cases, urine amylase is normal or low.

Custom Additional Information

Causes of high serum amylase include acute pancreatitis, pancreatic pseudocyst, pancreatic ascites, pancreatic abscess, neoplasm in or adjacent to pancreas, trauma to pancreas, and common duct stones.

Nonpancreatic causes of hyperamylasemia include inflammatory salivary lesions (eg, mumps), perforated peptic ulcer involving pancreas or not, intestinal obstruction and infarction, afferent loop syndrome, biliary tract disease including stones, aortic aneurysm, peritonitis, acute appendicitis, cerebral trauma, burns and traumatic shock, the postoperative state (with and without pancreatitis), diabetic ketoacidosis, and extrapancreatic carcinomas (especially of esophagus, lung, ovary). Amylase levels more than 25-fold the upper limit of normal are often found when metastatic tumors produce ectopic amylase. Such levels are higher than those usually found in cases of pancreatitis.³ In renal insufficiency amylase is usually not more than three times the upper limit of normal. Moderate increases may be reported in normal pregnancy. Increases may be found with tubo-ovarian abscess, ruptured ectopic pregnancy, macroamylasemia, and with a substantial number of drugs, including morphine. Relationships between pancreatitis and hyperlipidemias types I, IV, and V are described. Amylasemia may be associated with hyperparathyroidism.

Macroamylase is a high molecular weight material, normal amylase complexed to high molecular weight protein such as immunoglobulin. It is characterized by high serum amylase and low to normal urine amylase. Macroamylase occurs in normal as well as abnormal subjects.⁴

Other tests: In **pancreatitis**, varying percentages of patients have the following other abnormalities in varying combinations: elevation of triglyceride, alkaline phosphatase, AST (SGOT), total bilirubin, white blood cell count, left shift. Calcium levels should be followed in fulminant pancreatitis, since extremely low serum calcium levels can evolve. **Serum lipase and two-hour urine amylase** may both be extremely valuable. Although determination of serum methemalbumin has been advocated as a test for acute hemorrhagic pancreatitis, it is cumbersome and is not done in many American laboratories.

Isoenzymes of amylase exist: pancreatic and salivary type, as noted under Limitations. They can be separated by polyacrylamide gel or agarose film electrophoresis, isoelectric focusing, ion exchange chromatography, and plant isoamylase inhibitors. A monoclonal antibody approach is described.^{3,5} Amylase isoenzymes are separated in few laboratories. Where available the procedure is an expensive one. It is useful in assessing the decrease of pancreatic function in cystic fibrosis, in children older than five years, who may be candidates for enzyme replacement.

Specimen Requirements

Specimen

Serum (preferred) **or** plasma

Volume

1 mL

Minimum Volume

0.7 mL (**Note:** This volume does **not** allow for repeat testing.)

Container

Red-top tube, gel-barrier tube, **or** green-top (lithium heparin) tube. Do **not** use oxalate, EDTA, or citrate plasma.

Collection Instructions

Separate serum or plasma from cells within 45 minutes of collection.

Stability Requirements

Temperature	Period
Room temperature	14 days
Refrigerated	14 days
Frozen	14 days
Freeze/thaw cycles	Stable x3

Reference Range

See table.

Age	Male (U/L)	Female (U/L)
0 to 30 d	22–92	22–92
31 d to 6 m	8–71	8–71
7 m to 1 y	24–98	24–98
>1 y	31–110	31–110
*LabCorp internal studies		

Storage Instructions

Maintain specimen at room temperature.

Causes for Rejection

Improper labeling

References

Science and Technology, LabCorp, Burlington, NC for reference intervals.

Footnotes

1. McNeely MD. Pancreatic function. In: Kaplan LA, Pesce AJ, eds. *Clinical Chemistry Theory, Analysis, and Correlation*. 2nd ed. St Louis, Mo: Mosby-Year Book Inc;1989: 390-397.
2. Ellis C, Koehler DF, Eckfeldt JH, Levitt MD. Evaluation of an inhibitor assay to determine serum isoamylase distribution. *Dig Dis Sci*. 1982 Oct; 27(10):897-901. PubMed 6180877 (<http://www.ncbi.nlm.nih.gov/pubmed/6180877>)
3. Eckfeldt JH, Levitt MD. Diagnostic enzymes for pancreatic disease. *Clin Lab Med*. 1989 Dec; 9(4):731-743. PubMed 2480201 (<http://www.ncbi.nlm.nih.gov/pubmed/2480201>)
4. Van Gossum A. Macroamylasemia: A biochemical or clinical problem? *Dig Dis*. 1989; 7(1):19-27. PubMed 2466587 (<http://www.ncbi.nlm.nih.gov/pubmed/2466587>)
5. Warshaw AL, Hawbolt MM. Puzzling persistent hyperamylasemia, probably neither pancreatic nor pathologic. *Am J Surg*. 1988 Mar; 155(3):453-456. PubMed 2449825 (<http://www.ncbi.nlm.nih.gov/pubmed/2449825>)